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Website: <https://appliedcryptography.page>

Quiz 1.6: Collision Resistant Hash Functions

This completely optional quick quiz acts as a learning aid to help you find out how much you absorbed from our Collision Resistant Hash Functions session.^a Remember, this is just for fun and to help you identify areas you might want to review. Each question has three possible answers, but only one is correct. Take your time, and when you're done, check your answers against the answer key at the end. Good luck!

^a<https://appliedcryptography.page/slides/#1-6>

Questions

1. What is a collision in the context of hash functions?
 - A. When two different inputs produce the same hash output
 - B. When a hash function takes too long to compute
 - C. When the hash output is larger than the input
2. According to the birthday paradox, how many people do you need for a 50% chance of a shared birthday?
 - A. 23 people
 - B. 50 people
 - C. 183 people
3. What is the purpose of adding salt to password hashing?
 - A. To compress the password before hashing
 - B. To prevent precomputation attacks like rainbow tables
 - C. To make the hash computation faster
4. Which hash function construction is vulnerable to length extension attacks?
 - A. Sponge construction (used in SHA-3)
 - B. HMAC construction
 - C. Merkle-Damgård (used in SHA-2)
5. What researcher demonstrated the first practical MD5 collision in 2004?
 - A. Xiaoyun Wang
 - B. Colin Percival
 - C. Bruce Schneier
6. What is HMAC designed to protect against?
 - A. Collision attacks
 - B. Length extension attacks
 - C. Birthday attacks

7. In the sponge construction, what prevents length extension attacks?
 - A. Using a different compression function
 - B. Processing blocks in parallel instead of sequentially
 - C. Hiding part of the internal state (capacity) from the output
8. What was the name of the 2017 SHA-1 collision attack by Google?
 - A. Heartbleed
 - B. POODLE
 - C. SHAttered
9. Why is storing passwords in plaintext dangerous?
 - A. Attackers immediately get all passwords if the server is compromised
 - B. It uses too much storage space
 - C. It makes login verification slower
10. What makes Scrypt a memory-hard function?
 - A. It requires storing a large array of pseudorandom values during computation
 - B. It encrypts the password multiple times
 - C. It uses very large hash outputs
11. Approximately how many SHA-256 hashes can a high-end GPU compute per second?
 - A. 2+ billion hashes/second
 - B. 20,000 hashes/second
 - C. 1 million hashes/second
12. What is PBKDF2?
 - A. A key derivation function that iterates a PRF many times
 - B. A type of rainbow table
 - C. A memory-hard hash function like Scrypt
13. What property must collisions have due to the pigeonhole principle?
 - A. They are impossible to find
 - B. They must exist when infinite inputs map to finite outputs
 - C. They only occur with weak hash functions
14. What is the Random Oracle Model?
 - A. A type of attack on hash functions
 - B. A real hash function implementation
 - C. A theoretical framework modeling hash functions as truly random functions
15. Why is encrypting passwords (instead of hashing) a bad idea?
 - A. Encrypted passwords take up too much storage space
 - B. Encryption is slower than hashing
 - C. The server needs the decryption key, which an attacker could also steal
16. What makes rainbow tables ineffective against salted hashes?
 - A. Each unique salt requires a completely new rainbow table
 - B. Salted hashes are computed faster
 - C. Salt makes the hash output longer
17. What is protocol agility in cryptography?
 - A. A type of timing attack on protocols

- B. The speed at which protocols can be implemented
 - C. The ability to support and transition between multiple cryptographic algorithms
18. How much computational effort did the SHattered attack require?
- A. About 6,500 CPU years and 110 GPU years
 - B. About 1 CPU year
 - C. About 1 million CPU years
19. What discovery about Joux's multi-collision attack was surprising?
- A. Finding 2^n collisions only takes about n times the work of finding one collision
 - B. Multi-collisions are impossible to find
 - C. Finding multi-collisions takes exponentially more work
20. Why does Bitcoin mining demonstrate hash function security?
- A. It proves that hash collisions don't exist
 - B. The extreme computational cost shows how hard it is to find specific hash patterns
 - C. It uses quantum-resistant hash functions

Answers

Question	Key
1	A
2	A
3	B
4	C
5	A
6	B
7	C
8	C
9	A
10	A
11	A
12	A
13	B
14	C
15	C
16	A
17	C
18	A
19	A
20	B